INSTRUMENT CLUSTER (Continued)

the Off position battery current is not supplied to any gauges, and the EMIC circuitry is programmed to move all of the gauge needles back to the low end of their respective scales. Therefore, the gauges do not accurately indicate any vehicle condition unless the ignition switch is in the On or Start positions.

All of the EMIC gauges are air core magnetic units. Two fixed electromagnetic coils are located within each gauge. These coils are wrapped at right angles to each other around a movable permanent magnet. The movable magnet is suspended within the coils on one end of a pivot shaft, while the gauge needle is attached to the other end of the shaft. One of the coils has a fixed current flowing through it to maintain a constant magnetic field strength. Current flow through the second coil changes, which causes changes in its magnetic field strength. The current flowing through the second coil is changed by the EMIC circuitry in response to messages received over the PCI data bus. The gauge needle moves as the movable permanent magnet aligns itself to the changing magnetic fields created around it by the electromagnets.

The gauges are diagnosed using the EMIC self-diagnostic actuator test. (Refer to 8 - ELECTRICAL/ INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). Proper testing of the PCI data bus and the electronic data bus message inputs to the EMIC that control each gauge require the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information. Specific operation details for each gauge may be found elsewhere in this service information.

VACUUM-FLUORESCENT DISPLAY

The Vacuum-Fluorescent Display (VFD) unit is soldered to the EMIC electronic circuit board. With the ignition switch in the Off or Accessory positions, the odometer display is activated when either door is opened (Rental Car mode) and is deactivated after both doors are closed. If a door is left open with the ignition switch in any position except On or Start, the VFD will remain illuminated until the interior lights control battery saver (load shedding) timer expires after about twenty minutes. Otherwise, the display unit is active when the ignition switch is in the On or Start positions, and inactive when the ignition switch is in the Off or Accessory positions.

The illumination intensity of the VFD unit is controlled by the EMIC circuitry based upon an input from the headlamp switch circuitry and a dimming level input received from the panel lamp dimmer switch circuitry. The EMIC synchronizes the illumination intensity of other VFD units with that of the unit in the EMIC by sending electronic dimming level messages to other electronic modules in the vehicle over the PCI data bus.

The EMIC VFD unit has several display capabilities including odometer, trip odometer, software version, and can display various diagnostic information when certain fault conditions exist. An odometer/trip odometer switch on the EMIC circuit board is used to control some of the display modes. This switch is actuated manually by depressing the odometer/trip odometer switch button that extends through the lower edge of the cluster lens, just right of center. Actuating this switch momentarily with the ignition switch in the On position will toggle the VFD between the odometer and trip odometer modes. Depressing the switch button for about two seconds while the VFD is in the trip odometer mode will reset the trip odometer value to zero. Holding this switch depressed while turning the ignition switch from the Off position to the On position will initiate the EMIC self-diagnostic actuator test. Refer to the appropriate diagnostic information for additional details on this VFD function. The EMIC microprocessor remembers which display mode is active when the ignition switch is turned to the Off position, and returns the VFD display to that mode when the ignition switch is turned On again.

The VFD is diagnosed using the EMIC self-diagnostic actuator test. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). Proper testing of the PCI data bus and the electronic data bus message inputs to the EMIC that control some of the VFD functions requires the use of a DRBIII® scan tool. Refer to the appropriate diagnostic information. Specific operation details for the odometer and trip odometer functions of the VFD may be found elsewhere in this service information.

INDICATORS

Indicators are located in various positions within the EMIC and are all connected to the EMIC electronic circuit board. The antilock brake system indicator, headlamp high beam indicator, fog lamp indicator, and turn signal indicators operate based upon hard wired inputs to the EMIC. The brake indicator is controlled by the EMIC programming and hard wired park brake and brake warning indicator switch inputs to the EMIC. In vehicles without the optional off-road package the four-wheel drive indicator is controlled by a hard wired input from the transfer case switch, while in vehicles equipped with the off-road package this indicator is controlled by PCI data bus messages from the Powertrain Control Module (PCM). The rear and front lock indicators are controlled by the EMIC programming and hard wired inputs from the axle locker switch and the rear/front locker indicator switches. The seatbelt indicator is controlled by the EMIC programming and a hard wired seat belt switch input to the EMIC. The Mal-